Line PCB

A) Power: Power PCB → this PCB rails (5V for Teensy, 3.3V for sensors)

1. Power input to this PCB (from Power PCB)
   Power PCB 5V OUT → This PCB 5V_RAIL
   Power PCB GND → This PCB GND_RAIL
   Power PCB 3.3V OUT → This PCB 3.3V_SENS_RAIL
   Power PCB GND → This PCB GND_RAIL (common ground everywhere)

2. Teensy power (on this PCB)
   5V_RAIL → Teensy **VIN**
   GND_RAIL → Teensy **GND**

3. Bulk capacitor on sensor 3.3V rail (on this PCB)
   Capacitor **C1 = 100µF/25V electrolytic**
   C1 **+** → **3.3V_SENS_RAIL**
   C1 **−** → **GND_RAIL**

B) QTR-MD-05A Sensors (4 boards) power + decoupling + control

For EACH QTR board (QTR1, QTR2, QTR3, QTR4):

1. Power
   QTRx **VCC** → **3.3V_SENS_RAIL**
   QTRx **GND** → **GND_RAIL**
2. Ceramic capacitor close to the sensor connector (on this PCB)
   Capacitor **C4/C5/C6/C7 = 0.1µF (100nF) ceramic**
   Cap **one side** → **3.3V_SENS_RAIL**
   Cap **other side** → **GND_RAIL**
3. Emitter control
   QTRx **CTRL_ODD** → Teensy **pin D6** (shared line to all 4 boards)
   QTRx **CTRL_EVEN** → **NC**

C) Analog outputs wiring (18 used, 2 not connected)

QTR #1 outputs → Teensy analog pins
QTR1 OUT0 → Teensy **A0**
QTR1 OUT1 → Teensy **A1**
QTR1 OUT2 → Teensy **A2**
QTR1 OUT3 → Teensy **A3**
QTR1 OUT4 → Teensy **A4**

QTR #2 outputs → Teensy analog pins
QTR2 OUT0 → Teensy **A5**
QTR2 OUT1 → Teensy **A6**
QTR2 OUT2 → Teensy **A7**
QTR2 OUT3 → Teensy **A8**
QTR2 OUT4 → Teensy **A9**

QTR #3 outputs → Teensy analog pins
QTR3 OUT0 → Teensy **A10**
QTR3 OUT1 → Teensy **A11**
QTR3 OUT2 → Teensy **A12**
QTR3 OUT3 → Teensy **A13**
QTR3 OUT4 → Teensy **A14**

QTR #4 outputs → Teensy analog pins (3 used, 2 dropped)
QTR4 OUT0 → **NC**
QTR4 OUT1 → Teensy **A15**
QTR4 OUT2 → Teensy **A16**
QTR4 OUT3 → Teensy **A17**
QTR4 OUT4 → **NC**

D) UART pins
Teensy **TX7** → Main PCB **RX**
Teensy **RX7** → Main PCB **TX**